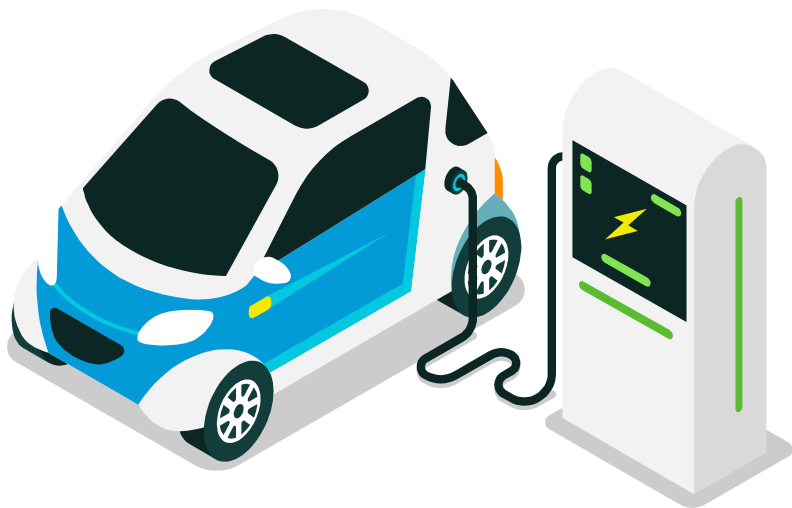




Assessing Public Opinion on Electric Vehicles Using Twitter Data

Katayoon Kaviani
Data Scientist, Tesla

Overview of the project



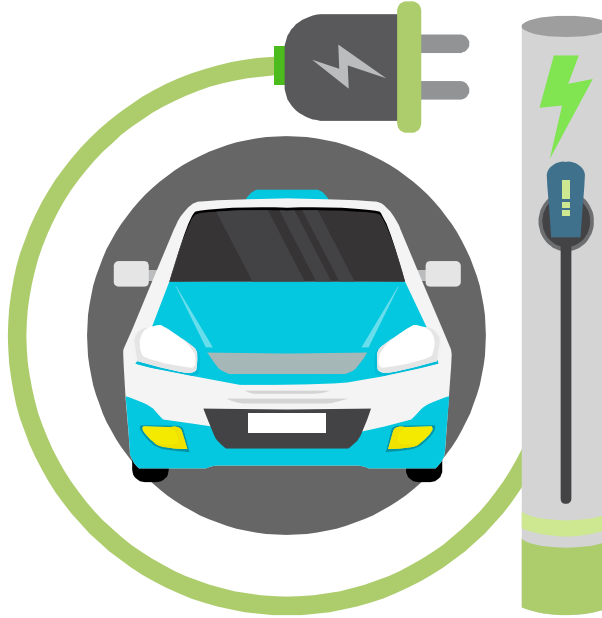
Data Scraping



Data Cleaning and
Preprocessing



Sentiment Analysis



Exploratory Data
Analysis



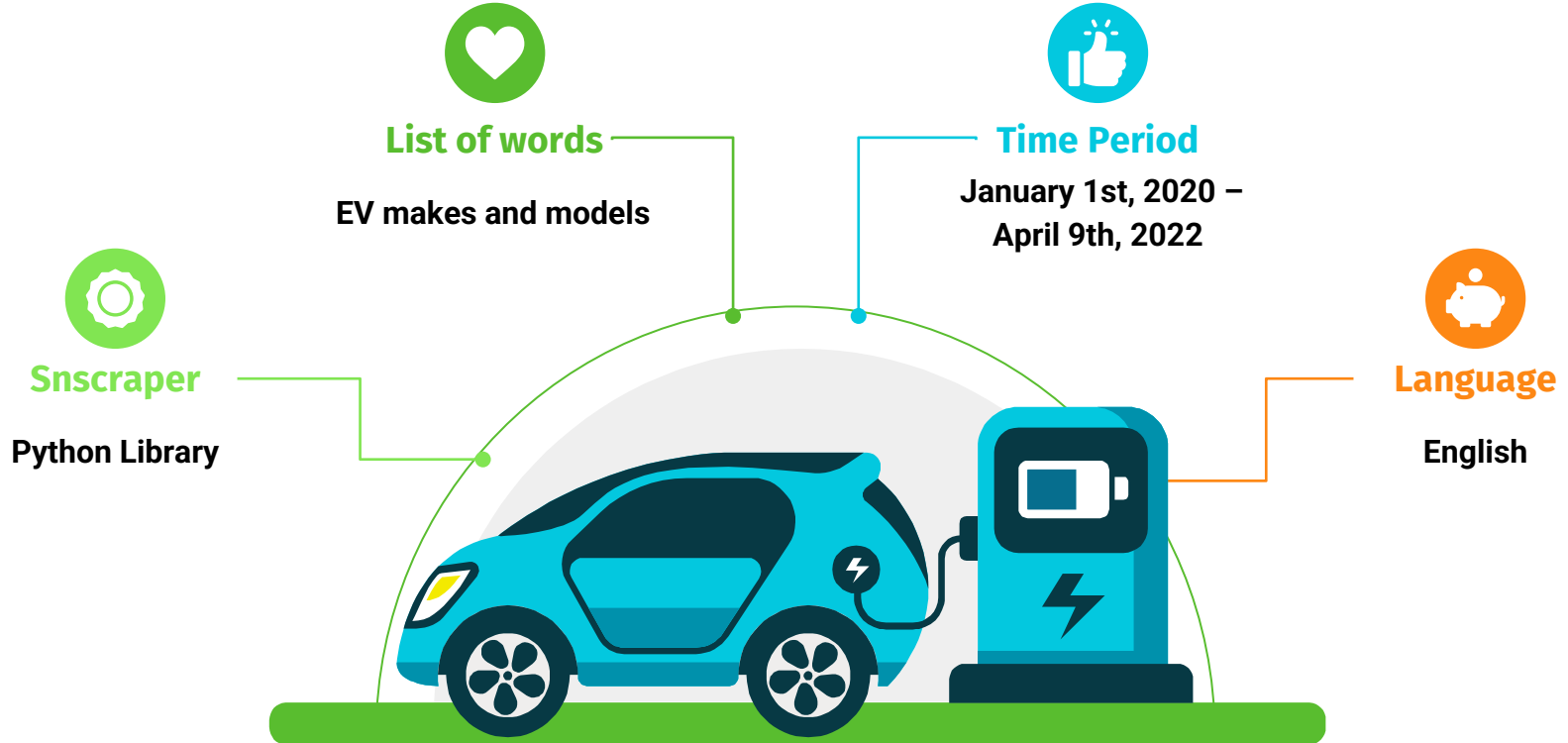
Topic Modeling



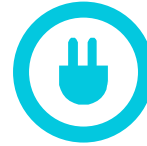
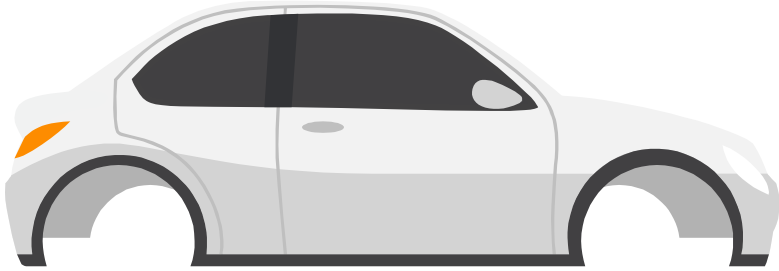
Insights and Future
Work



Data Collection and Scraping



Data Cleaning and Preprocessing



Removnig usernames and links



Removing tweets without relevant words in them



Removing Advertisemnets by hashtags

Sentiment Analysis



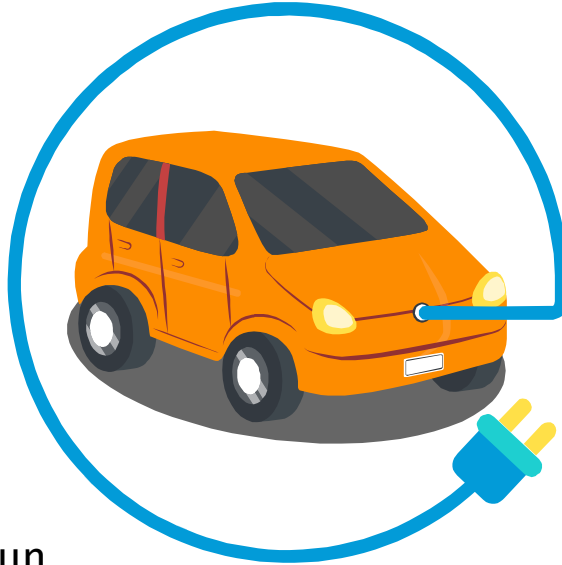
Hugging Face

- Offers open-source libraries and tools for natural language processing (NLP) tasks.
- Libraries are widely used by researchers, developers, and data scientists to build and train state-of-the-art NLP models.
- Offers several pre-trained models for various NLP tasks, including sentiment analysis.
- One of the models used for this project is the Twitter RoBERTa Base Sentiment model from CardiffNLP, which is trained on Twitter data for sentiment analysis.



Sentiment Analysis

- Labels were defined as Negative, Neutral, and Positive
Compound scores were calculated afterwards

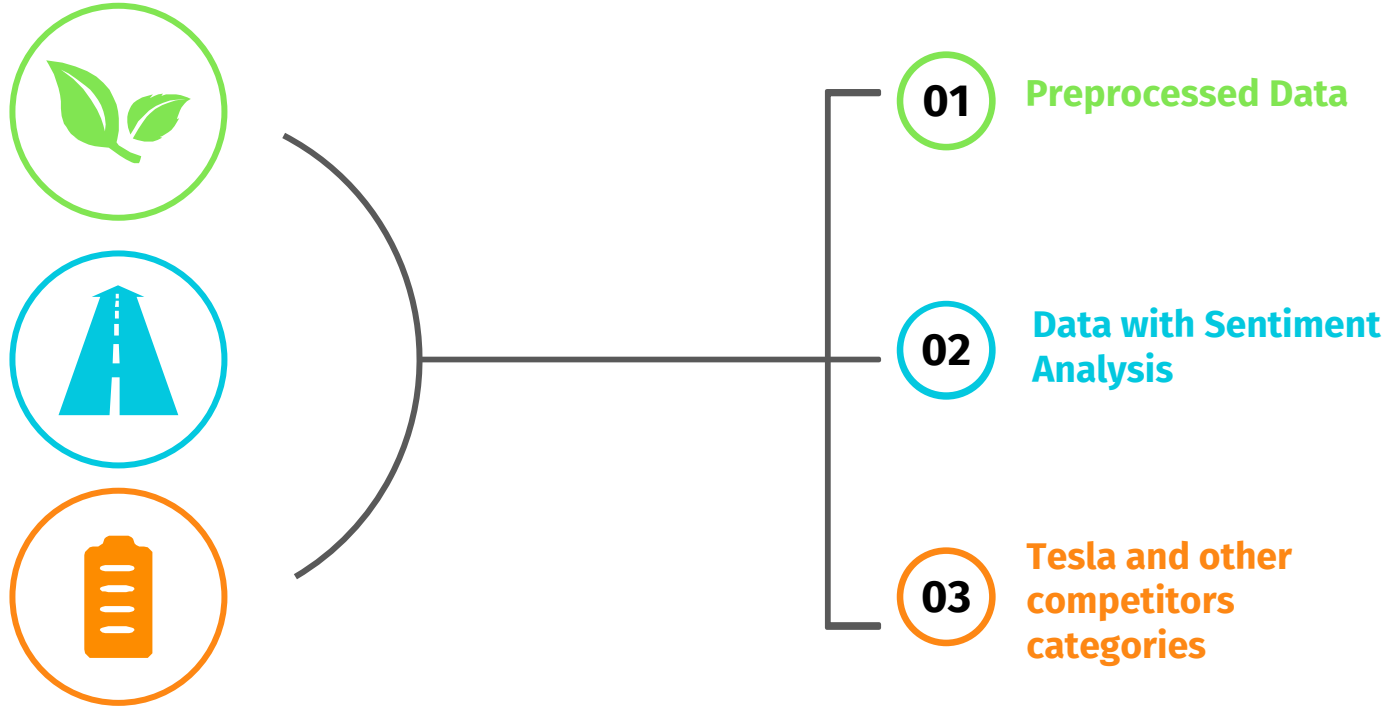


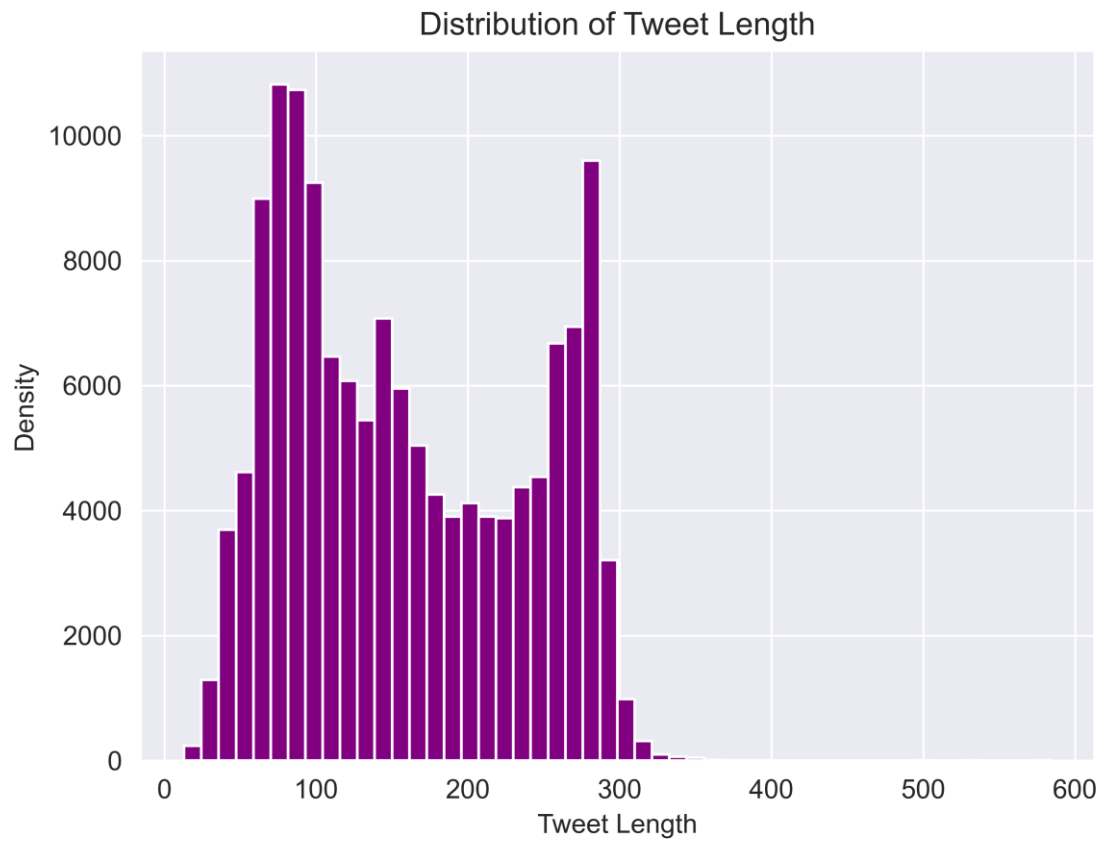
- TensorFlow was used to run the model and get the sentiment scores

- **TFAutoModelForSequenceClassification** a pre-trained deep learning model

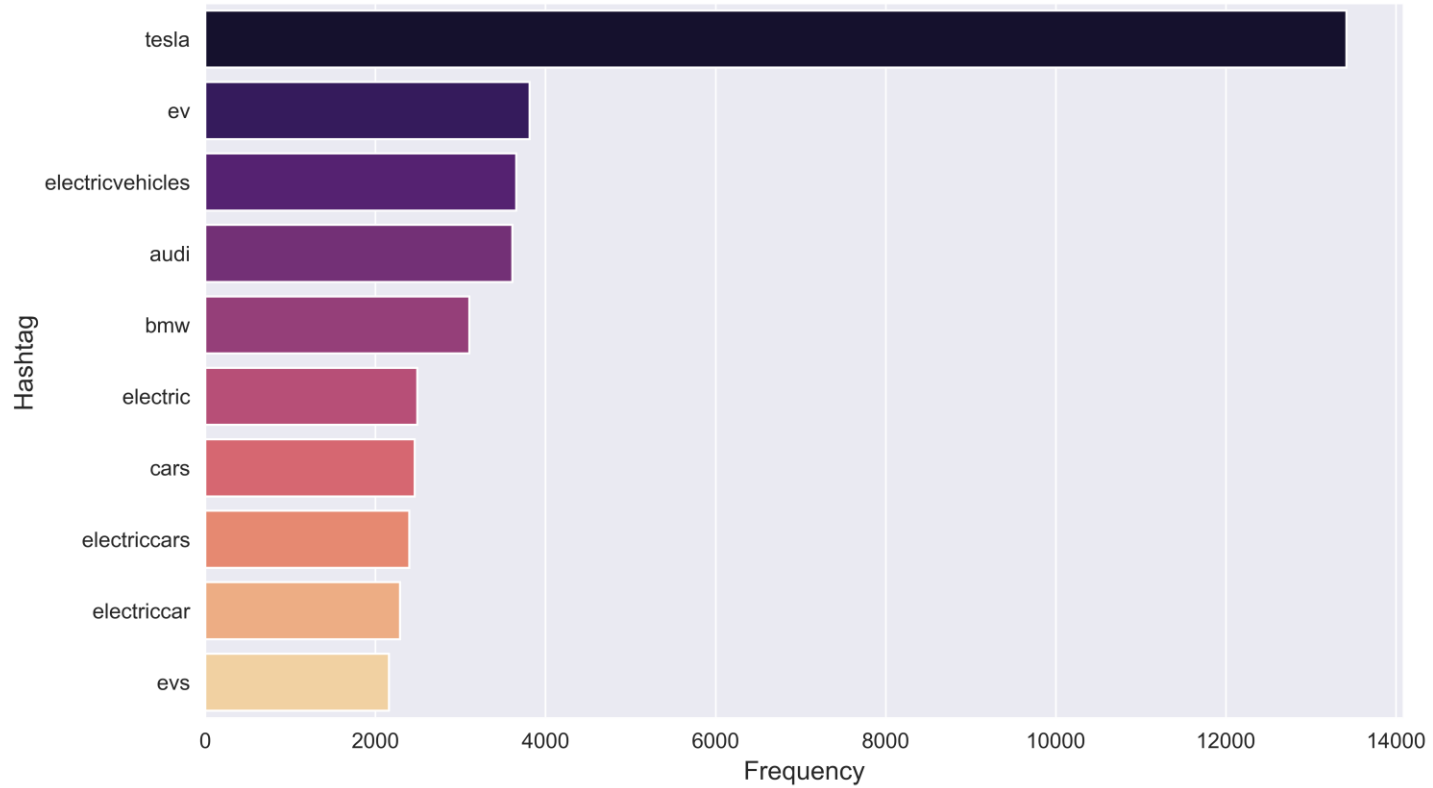
- Range of compound scores: $(-1, 1)$
- Range of scores: $(0, 1)$

Exploratory Data Analysis

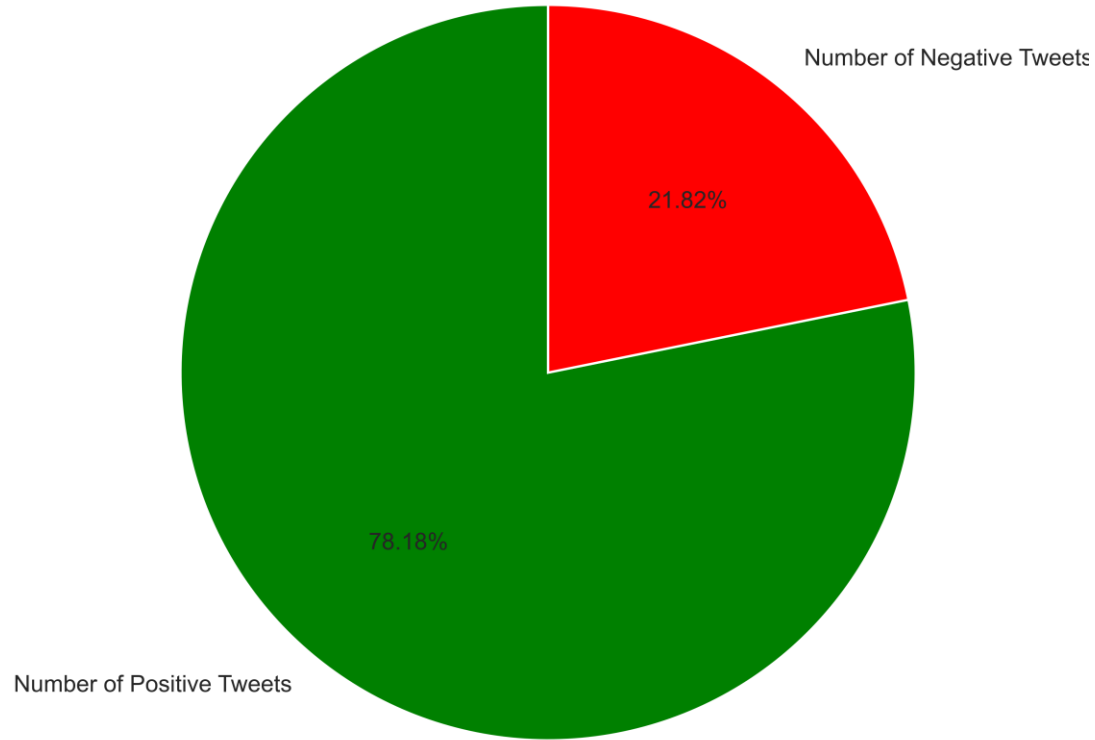




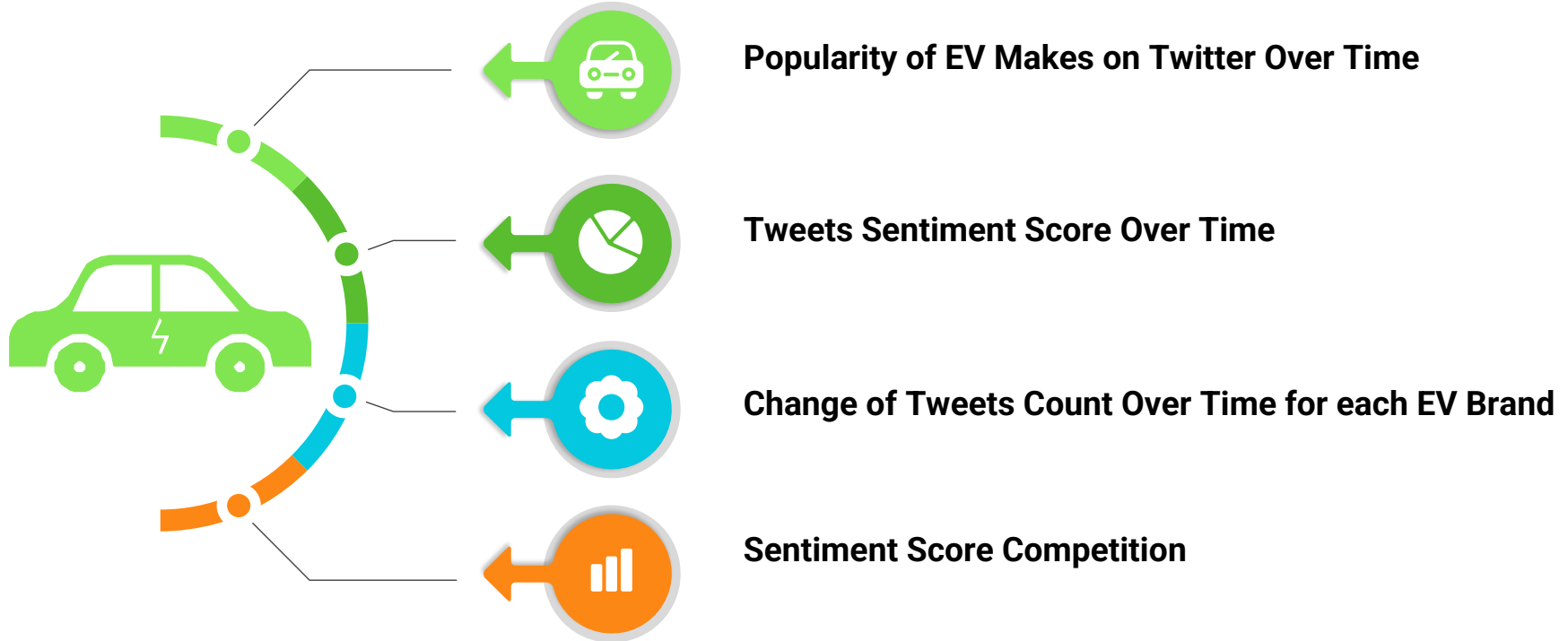
Top 10 Hashtags



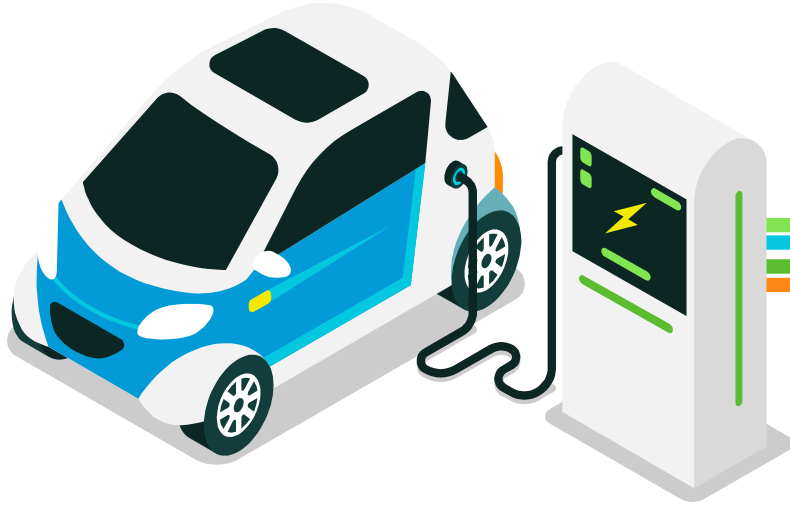
Positive tweets vs. Negative tweets



Tweets Analysis Over Time



Topic Modeling



Latent Dirichlet Allocation (LDA)



**Countvectorizer:
Transform text to vector**



**TfidfVectorizer :
Word frequency in tweets**

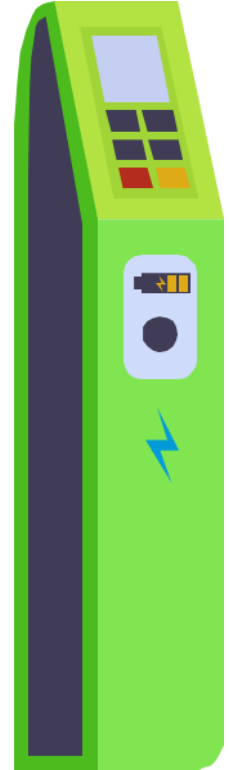


Find the optimal LDA model

Topic Modeling

Let's take a look at topic modeling results:

- BMW
- Mercedes-Benz
- Tesla



Competitors Insights...

- The Model X is not a good SUV. R1S runs laps around it.
- I had a Model S before having children. It's not easy parking it between the lines and then having to open the long doors. especially in the city. Model X Plaid will replace our Model Y after we take delivery of the Rivian R1S that will serve as our grocery mule.
- It's more about Model X vs R1x. If you just need a basic crossover, grab a Model Y and save \$30k. Model S is somewhat played out - a Taycan is a better all around car, it just doesn't have the same 1000+HP party trick. Lucid is stealing customers who want the max range.

Competitors Insights...

- Definitely. R1S is a much better product than a Glorified Van(model X).
- I have no doubt that Lucid Air is going to impact Model S sales and Rivian R1S will impact X sales. In summary, I 😊 model S (🤨 3, X, Y). But I would not trust FSD on surface streets. Not at all. The technology is not there yet. So consider those extra \$7K. Though it will be like \$10K after purchase. Maybe your year will be lucky to include all the reqs... Good Luck.
- What year model is yours the 2021 I was in was extremely quiet. Some of the things you stated were actually things I did not like with the Rivian. Road noise and ride, also the seats were very uncomfortable compared to the X and our Lexus. They have also had service issues.

Tesla Insights...

- The order rate for the Model Y peaked a couple days after the price cuts in January and declined over the remainder of the quarter ending March below the orders levels in Q4. The Model 3 is a disaster. The S/X are dead.
- EQS SUV is pretty close to Model X as well. I know quite a few who ended up buying X because they don't like S and Y is too small for family.
- The spread between model 3/Y and Model S/X was too wide for near identical product. That's the main issue. Y selling like pancakes why buy something double the price
- The Model X plaid brake color doesn't match the UltraRed. Would be great to update the brake paint. In day light, the mismatch is quite obvious.
- Elon, would you consider making an SUV version of the Cybertruck for the US market? While great, the Model Y is sadly too small for a lot of Americans and the Model X is too expensive.'

Tesla Insights...

- I have been having issues with my Model X's climate control and key fobs for over a year. I have been in service 6 times with no fix. Is there any way we can escalate this and find a resolution? I love my car and just want it to work properly!
- Is there an option to not have plastic around the wheel arches on the model X? It looks awful.
- Weird how so many model X windshield seem to crack. It almost like the body has too much flex in it.
- I would buy one right now if they would make a decent car for large families. So more than 5 seats. Unfortunately the Model Y 7 seater is a bit small and not available in Europe. And the Model X is too expensive.
- Tesla has to focus on technology because they seem to struggle with mechanical 'luxury' features like sunroofs that actually open, model S door handles, model X falcon wing. Cousin's boss bought a X last month, already in the shop door issues

Tesla Insights...

- The order rate for the Model Y peaked a couple days after the price cuts in January and declined over the remainder of the quarter ending March below the orders levels in Q4. The Model 3 is a disaster. The S/X are dead.
- Had 2 years of absolute hell with my model X. The quality from day one was terrible I should have rejected it. Been to 3 different service centers to try and find decent service and that's impossible. The car has constant issues. I WILL NEVER buy this brand ever again.
- My Model X, in Chicago winter, only gets ~130 miles on a charge.... And that's w/o heat running high either. (~67F speed 4 fan / driver heated seat)

And looking through a lot more positive and negative tweets brought us the following result..

Tesla Insights

Software

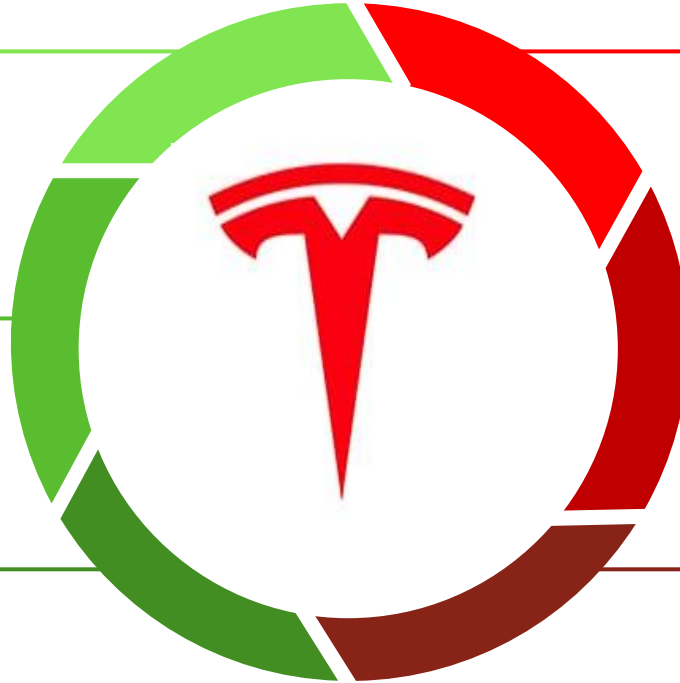
Build

Model 3/Y Price

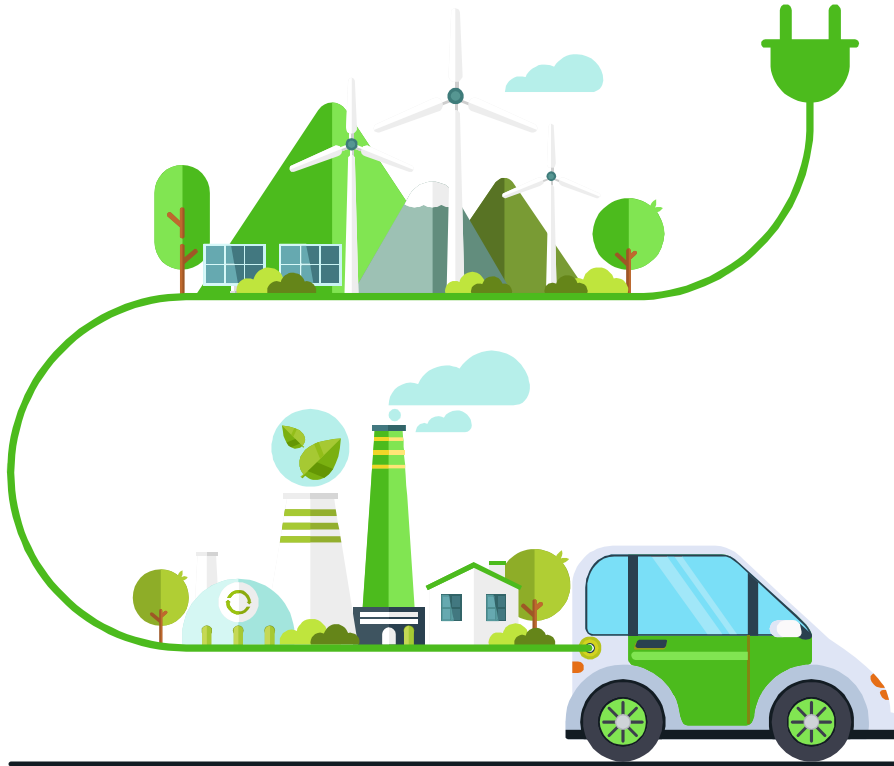
No Family Size

Battery

Autopilot



Future Work



- More Filtering on Tweets to get the most related tweets
- Removing Ads
- Improving Sentiment Analysis on comparison sentences
- Trying other models and algorithms
- Scraping more tweets
- Considering more competitors
- Prediction of future sentiment scores